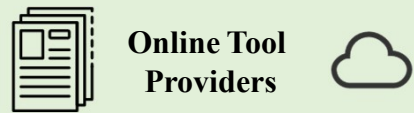
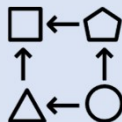


STAGE 1: RAW DATA & TOOL SOURCES



STAGE 2: TOOL CURATION & EXECUTABILITY FILTERING

Rule-based Tool Filtering & Executability Check



- ✓ Interface validity & parameter completeness checks
- ✓ Deduplication and basic constraint validation
- ✓ Executability check

Validated Tool Inventory

STAGE 3: TOOL NORMALIZATION & MANIFEST CONSTRUCTION

Automated Tool Manifest Normalization



Unified signatures



Canonical arguments



Output schema alignment

STAGE 4: FINANCE ATTRIBUTE ANNOTATION

LLM-based Finance Attribute Annotation

Update Frequency

Intent Type

Regulatory Domain

Sample schema:

```
{
  "update_frequency": xxx,
  "intent_type": xxx,
}
```

STAGE 5: QUESTION SOURCE & QUESTION SELECTION

Finance QA Sources

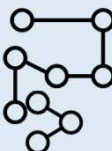
Tool-required Question Filtering



STAGE 6: TOOL-QUESTION ALIGNMENT & VERIFICATION

Semantic Tool Retrieval & Executability

- Top-K semantic retrieval
- Multi-sample
- Execution check



STAGE 7: HUMAN-IN-THE-LOOP STAGE VERIFICATION



Human Expert Review & Quality Assurance

- Annotation spot-checking
- Tool-question alignment verification
- Compliance & format validation

STAGE 8: FINAL DATASET OUTPUT

Benchmark Tool Library

Benchmark Question Set

Runnable Benchmark